

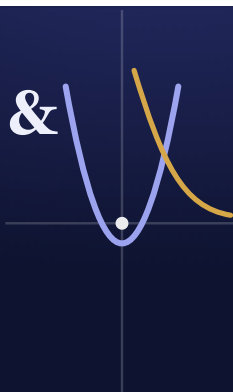
CLASS SYLLABUS · 2026-2027

Engineering Design & Problem Solving

Solve a real problem, design it, and present it like an engineer.

Mr. Gavaldon · Da Vinci School

Contact: robertgavaldon@burnhamwood.org



DAILY How this class runs

- 1 Grab your **assigned laptop**
- 2 **Sign in** to the program we're using
- 3 **Start work** — pick up where you left off
- 4 **Wait for instruction** when I call the class

PHONES

Up at the **start, every day**. Like the movies: phone away so everyone can focus. **I'll ask once** — then an email home, then a parent + admin meeting. Goal: **never** need it.

TUTORING & NOT FAILING

Fail a test → **one week of mandatory tutoring** (the department or me, on tutoring days). It's how we keep you from failing. Seats are assigned.

QUALITY OF WORK

- ✓ Finish your work **on your own**.
- ✓ Classwork done **in class**, the day it's assigned.
- ✓ On THIS class? No other classes or activities.
- ✓ **No AI in class or on tests** — you have to know it.

NOT ACCEPTED

- ✗ Yelling, fighting, hitting/touching others
- ✗ Eating, cussing, out of uniform
- ✗ Not turning in / not working in class
- ✗ No-showing tutoring; table-hopping; daily restroom

CORE 4

Respect

Responsibility

Quality of Work

Quality of Self

Effort you can see.

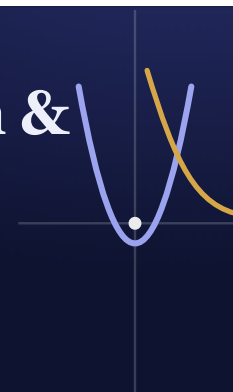
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Course Roadmap

Engineering Design & Problem Solving
tentative – may shift

SEMESTER 1 · FALL

1 6-Wks
Aug – Sep

Foundations & Great Presentations

5w

Notebook, safety, units, and what makes a presentation land

2 6-Wks
Oct – Nov

Design Process + Fusion

5w

Full design process on a real problem; CAD to communicate a design

3 6-Wks
Nov – Dec

Team Design Project 1 + Pitch

5w

Team roles, build, and a formal design presentation

★ Semester 1 Final

SEMESTER 2 · SPRING

4 6-Wks
Jan – Feb

InSPIRESS — Design the Payload

6w

Define the problem and model a competition payload (InSPIRESS / UAH)

5 6-Wks
Mar – Apr

Build, Test & Iterate

6w

Prototype, test against requirements, and improve the design

6 6-Wks
Apr – May

Final Presentation & Portfolio

6w

Competition-grade presentation + senior design portfolio

★ Final

BRING EVERY DAY

- ✓ A pencil or pen
- ✓ A pencil sharpener
- ✓ A glue stick
- ✓ A folder just for this class
- ✓ A composition notebook
- ✓ Colored pencils or markers
- ✓ Scissors
- ✓ A mouse

For the class — bring one: hand sanitizer, a box of Kleenex, or disinfecting wipes.

iPads are allowed for class work. The first time I see texting, it goes away — and it does not come back out.

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PROGRAM OF STUDY

The 4-Year Engineering Pathway

CTE Levels 1-4 · Grades 9-12 ·
leading to certification

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Grade 9

Principles of Applied Engineering

INTRODUCTORY

\$127.781

Engineering design process, safety, and **Autodesk Fusion CAD from week 2** — project past project.

▲ Fusion part-modeling portfolio

◆ Starts ACU: Sketching · Part Modeling

2

Grade 10

Manufacturing Engineering Technology I

INTERMEDIATE

\$127.813

Design → make: 3D printing/CNC, quality, and mechanical/electrical & automation systems.

▲ Prototype-to-production build

◆ Adds Assembly · Drawings · OSHA-10 (opt.)

3

Grade 11

Engineering Design & Problem Solving · InSPIRESS

ADVANCED

\$127.785

Full design process on a real problem; **design a payload and compete** (InSPIRESS / UAH).

▲ InSPIRESS competition entry

◆ Sit the ACU — Fusion exam

4

Grade 12

Engineering Design & Presentation II

CAPSTONE

\$127.784

Capstone: advanced system design + presentation; lead a competition-grade design cycle.

▲ InSPIRESS capstone + portfolio

◆ ACU earned · stretch ACP / dual credit

CERTIFICATION

Autodesk Certified User — Fusion

Certiport · 35-40 Q · 50 min · pass 700/1000

Levels follow the Texas CTE Program of Study (Level 1 intro → Level 4 capstone). Course codes tentative — confirm with the campus CTE coordinator.

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